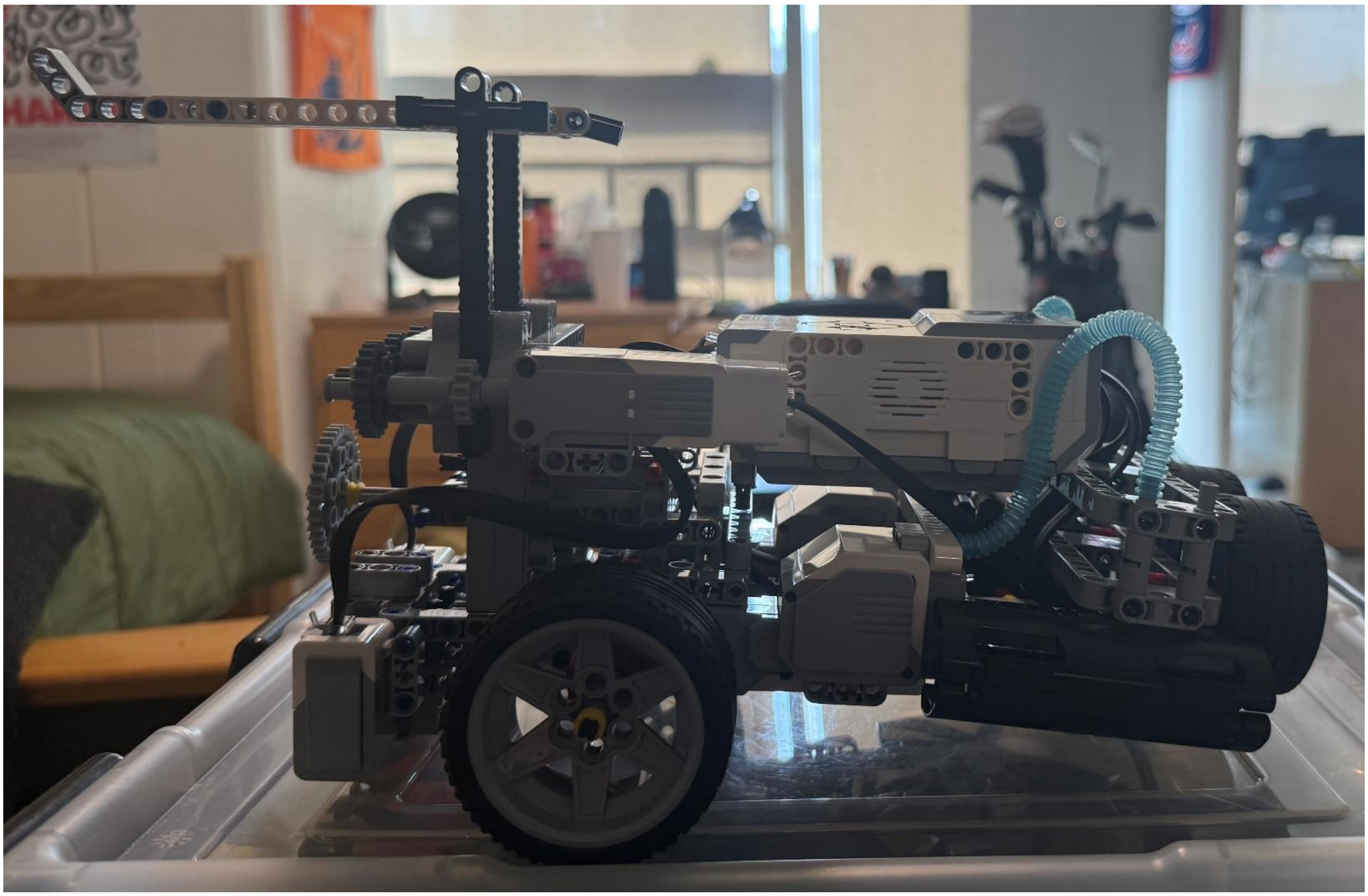
Project Robot

Group 1.15 - ENED 1002c-001

Started 1/12/2026 by: Cooper Hornaday, Dane Moore, Ben Eberhardt, Christoph Simms



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Meeting Minutes

2-3

Two Hour Meeting (hours total)

Participants: Ben (2), Christoph (2), Cooper (2), and Dane (2).

What We Worked On:

- Lego Part inventory
- Code of Cooperation

Big Decisions:

- Set expectations for each team member over the course of the semester.

Key Ideas:

- The code of cooperation will keep our group effective while in and outside of meetings.

Remember:

- Respect your teammates and cooperate on team tasks. In this project we will need to rely on each other to hold their own weight.

On Task, we have no current tasks

2-10

Two Hour Meeting (hours total)

Participants: Ben (4), Christoph (4), Cooper (4), and Dane (4).

What We Worked On:

- Introduction to using LabView
- Design Specifications for the project

Key Ideas:

- The introduction to using LabView will give our group fundamental information about Lego Mindstorm EV3.
- Test different sensors work (color sensor).
- The Design specification gave our group project management information which we can use throughout the semester.

Remember:

- Working in cooperation with your teammates on each task will make the entire project move smoothly.

2-17

Two Hour Meeting (hours total)

Participants: Ben (6), Christoph (6), Cooper (6), and Dane (6).

What We Worked On:

- Project subtask 1
- Creating a strong base for our motors and sensors

Big Decisions:

- **Size of wheels in the front and back of the robot**

Key Ideas:

- Understand what is asked of us in Project subtask 1
- **Propper installation of sensors will lead to a successful project.**

Remember:

- There are hundreds of pieces for a reason, so experiment with the pieces to determine which works best.

On Task, our robot will be prepared for Project Subtask 1 by 2/24

2-22

One Hour Meeting (hours total)

Participants: Ben (7), Christoph (7), Cooper (7), and Dane (7).

What We Worked On:

- Preparing our robot for project subtask 1

Big Decisions:

- **We added a second color sensor to track the curve of the line**

Key Ideas:

- Learning how to program our robot through LabView will help us with future subtasks
- **Connecting each color sensor to the left and right motor will help the robot turn when a different color is sensed**

Remember:

- Make sure all chords are properly connected to the EV3 so the sensors and motors work properly.

On Task, our group is ready for Project Subtask 1

2-17

Fifteen Minute Meeting (hours total)

Participants: Ben (7.25), Christoph (7.25), Cooper (7.25), and Dane (7.25).

What We Worked On:

- Conducting Project Subtask 1

Remember:

- To get full credit, our group must show up on time and continuously run from zone 1-6 with no resets.

On Task, we completed Project Subtask 1 (5/5)

3/3

Two Hour Meeting (hours total)

Participants: Ben (9.25), Christoph (9.25), Cooper (9.25), and Dane (9.25).

What We Worked On:

- Activity 8 Subtask Teaming Reflection
- Updating our design notebook

Key Ideas:

- We need to start looking at and thinking about Project Status Update 1 and updating our design notebook

Remember:

- Our Project Status Update 1 and Design notebook must be completed and submitted before we leave for spring break (3/13)
- Refer to the previous semester's design notebook rubric to get full credit

On Task, we will continue to work on our assignments on our 3/10 meeting

3/9

Two Hour Meeting (hours total)

Participants: Ben (11.25), Christoph (11.25), Cooper (11.25), and Dane (11.25).

What We Worked On:

- Status update one
- Presenting status update one
- Lift system mechanism
- Rework the base of the robot (extra space for sensors, contains chords, battery compartment)

Key Ideas:

- We need to add excess accessibility to our robot design

Remember:

- In addition to the new features/sensors, we need to make sure that the previous features/sensors work properly

On Task

3/24

Two Hour Meeting (hours total)

Participants: Ben (13.25), Christoph (13.25), Cooper (13.25), and Dane (13.25).

What We Worked On:

- Activity 10: Descriptive Statistics and Testing
- Preparing to test our robots performance
- Lifting mechanism for robot

Key Ideas:

- By analyzing data and plots of our robot, we can adjust the mechanism to give our desired output.

Remember:

- **Project Subtask 2 is next week**

On Task, we will have a meeting outside of class time to prepare for Project Subtask 2

3/29

Two Hour Meeting (hours total)

Participants: Ben (15.25), Christoph (15.25), Cooper (15.25), and Dane (15.25).

What We Worked On:

- Activity 10: Descriptive Statistics and Testing
- Preparing our robot for Status update 2
- Recording test results in Excel
- Constructing a box with a handle for our EV3 to pick up

Key Ideas:

- We need to make sure our testing is consistent, so our measurements have no errors.

Remember:

- **Accurate testing will help us make decisions to our robot to improve performance during future tasks.**

On Task

3/31

Two Hour Meeting (hours total)

Participants: Ben (17.25), Christoph (17.25), Cooper (17.25), and Dane (17.25).

What We Worked On:

- Activity 10: Descriptive Statistics and Testing

- Preparing our robot for Status update 2
- Recording test results in Excel
- Constructing a box with a handle for our EV3 to pick up

Key Ideas:

- We will need our lifting mechanism working for the final presentation.

Remember:

- **Keep the design notebook updated with all testing information and images.**

On Task – We will need to have a meeting to work on Status update 2

4/2

One and a Half Hour Meeting (hours total)

Participants: Ben (18.75), Christoph (18.75), Cooper (18.75), and Dane (18.75).

What We Worked On:

- Testing the lifting mechanism
- Testing the robot following a dashed line
- Putting together and filming Status Update 2

Remember:

- **Keep the design notebook updated with all testing information and images.**

On Task

4/7

Two Hour Meeting (hours total)

Participants: Ben (20.75), Christoph (20.75), Cooper (20.75), and Dane (20.75).

What We Worked On:

- Adjusting our EV3 to fit the Final Demo requirements
- Start to think about our final presentation

Key Ideas:

- We will read through all the FAQ's to understand the Final Demo as much as possible.

Remember:

- **Keep the design notebook updated with all information and images regarding the final demo.**

On Task – We will need to have a meeting to prepare ourselves for the final demo

4/9

Three Hour Meeting (hours total)

Participants: Ben (23.75), Christoph (23.75), Cooper (23.75), and Dane (23.75).

What We Worked On:

- Preparing our robot for the final demonstration
- Adjusting our color sensor to sense a sticky note on the track, which drops the box
- Constructing a new box and adjustments to the lifting mechanism

Key Ideas:

- We need make sure our robot can use all its skills for the final demonstration

Remember:

- **Keep the design notebook updated with all information and images regarding the final demo.**

On Task – We need to meet again before the final demo

4/12

Three Hour and Fifteen Minute Meeting (hours total)

Participants: Ben (27), Christoph (27), Cooper (27), and Dane (27).

What We Worked On:

- Preparing our robot for the final demonstration
- Adjusting our code to properly display the power on the EV3
- Adjusting our code to fit the guidelines of the final demonstration

Remember:

- **Keep the design notebook updated with all information and images regarding the final demo**

On Task – We are ready for the final demonstration

4/14

Thirty Minute meeting (hours total)

Participants: Ben (27.5), Christoph (27.5), Cooper (27.5), and Dane (27.5).

What We Worked On:

- Running through the final demonstration

Key Ideas:

- We made it through subtasks 1 and 2 and to the final demonstration

Remember:

- **Keep the design notebook updated with all information and images regarding the final demo**

On Task – We need to take apart the robot and prepare for the presentation

4/16

One Hour Meeting (hours total)

Participants: Ben (28.5), Christoph (28.5), Cooper (28.5), and Dane (28.5).

What We Worked On:

- Taking apart and sorting all of the Lego pieces

Key Ideas:

- We need to keep track of all the pieces in each bag in order to turn in our robot

On Task – We need meet to work on the presentation

4/22

One Hour Long Meeting (hours total)

Participants: Ben (29.5), Christoph (29.5), Cooper (29.5), and Dane (29.5).

What We Worked On:

- Writing our slides to prepare for our final demonstration
- Practicing each of our slides

Remember:

- **We will be asked questions by the class, so we will come prepared.**

On Task

Project Management

Project name: Lego Minds
Project team: 1.15

Task ID	Task	Start	finish	Responsible	Responsible	3-Feb	10-Feb	17-Feb	3-Mar	10-Mar	17-Mar	24-Mar	7-Apr	14-Apr	20-Apr	27-Apr
1	Code Of Cooperation	3-Feb	10-Feb	Cooper	100											
2	Lego Invetory	3-Feb	10-Feb	Ben	100											
3	EV3 with LabVIEW	10-Feb	17-Feb	Dane	100											
4	Design Specifications	10-Feb	17-Feb	Dane	100											
5	Navigation System	10-Feb	17-Feb	Cooper	100											
6	Subtask 1	17-Feb	3-Mar	Christoph	100											
7	Teaming & Subtask Reflection	3-Mar	10-Mar	Ben	100											
8	Status Update	10-Mar	17-Mar	Cooper	100											
9	Lift System	10-Mar	24-Apr	Ben	100											
10	Status Update	24-Mar	7-Apr	Christoph	100											
11	Determing Contents	7-Apr	14-Apr	Dane	100											
12	Final Deliverables	7-Apr	14-Apr	Cooper	100											
13	Final Demo	14-Apr	19-Apr	Dane	100											
14	Presentations	20-Apr	27-Apr	Ben	100											

Emphasize

Design Specifications

Learn Our Interests

Stakeholders:

(What's important to stakeholders)

(How this will impact design)

(Conflicts with other stakeholders)

- *Robotic Companies*
  - Will want us to have a reliable code that won't fail, sustainable design, and a long run time.
  - They might want the design to more focused on coding and easy access to parts for improvements/repairs, and lastly a bigger battery for running time.
  - No conflicts with other stakeholders.
- *Design Engineers / Students*
  - Does the design fit the buyer's demands.
  - The design objective will be given to the engineers, so they won't change it.
  - Conflicts with Environmental Agencies over limitations.
- *Government*
  - The engineers use there funds efficiently and come up with a product that completes the task needed.
  - This will allow the design to be the best possible while using the least amount of resources.
  - Conflicts with the nonprofit researchers over cost of making design more ecofriendly.
- *Nonprofit Researchers*
  - The design helps people and creates a better future.

    - Will allows for the engineers to do whatever they want as long as it fits the goal.
    - Conflicts with the government over purpose of design.
- *Environmental Agencies*
    - The design meets the proper environmental law requirements.
    - Will give the engineers limitations to make sure its eco-friendly.
    - Conflicts with the design engineers over limitations.

Define

<u>Design Specifications</u>

Learn Our Needs

Criteria, design specifications:

- Long EB3 battery life
  - How long does the battery run for, looking for at least a hour.
- Long run time
  - How long do the sensors function for looking for 30 minutes.
- Environmental Impact
  - How much carbon does it amit, looking for it to meet standard amounts under 30kg CO2 over a 3-year lifespan.
- Inexpensive Model
  - What is the cost of the final design looking for under 100$ in class cost.
- Efficient Code
  - What percentage of runs are successful, looking for 9/10 runs succeed.
- Portability
  - Must be small enough to carry around anything less than 2ftx2ft.
- Sensors Working
  - Must correctly detect desired points then trigger actions within a second.
- Driving Straight
  - Must be able to drive in a straight line with no help within a 10cm deviation.
- Turning
  - Must be able to turn up to 60 degrees smoothly within an inch precision.
- Track A Solid/Dashed Curved Line
  - Must be able to track a curved dashed or solid line of 60 degrees and within a inch precision.
- Be Able to Pick up Object

- Must be able to pick up an object that is 150 grams or less in weight.

- Determine Contents
  - Must be able to determine the contents once observing for 20 seconds.

Constraints:

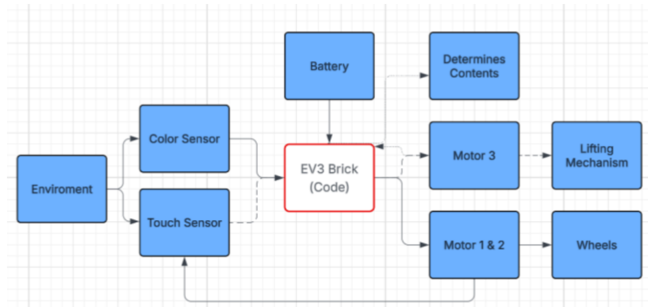
- Number Of Sensors
  - We only have a finite number of sensors so once we run out we must change the design.
- Port Slots Available
  - We only have four input slots limiting the sensors we are able to use and only 4 output port limiting the functions our robot can execute.
- Due Date / Time
  - We only have till Apr. 27th to complete the project causing us to have to hurry or rush the process.

Functions:

A: Major Functions

- Color Sensing – Will the robot be able to follow a color path or identify colors.
- Mobility – Will the robot be efficient moving and flow well or will it struggle to move.
- Efficiency of Code – When a action triggers the code will it execute fast or at all.
- Touch Sensing – Will the robot be able to do commands when needed due to it reaching / touching a desired point.
- Long battery life – Will the battery last enough to complete all the tasks needed.
- Design Sophistication – How easy is it to fix a problem with the robot and how many compartments are there.
- Lifting Object – Will the robot be able to lift objects on command or at trigger points.
- Determining Contents – Will the robot be able to determine the contents of a container when triggered/commanded.

B: Functional Block Diagram



Project Budget:

- We have four group members working for 2 hours each week for 15 weeks = 120 Hours
- We also are going to meet outside of class for 10 hours each person = 40 hours
- 40 + 120 = 160 total man hours = **32,000$**

Ideation

Needs (Solutions under prototyping):

- **Movement:** We needed to be able to move while only utilizing two motors.
    - We could use two wheels connected to motors in the front and a wheel in the back OR use a ball bearing in the back instead.
    - We could also just do two wheels in the back and two wheels in the front but then we do not have enough motors for that.
    - We could have two wheels in the front and only one in the back but when it goes to turn its going skirt around.

- **Turning:** We needed one motor to stop and the other to keep going.
    - We could have two wheels connected to one motor but then it wouldn't be able to turn well.
    - We also could have the wheels on two different motors in order to be able to stop one while the other one goes.
    - In order for the back to also turn smoothly we could put a ball bearing or another wheel unconnected to any motor.

- **Path Following:** We needed the robot to stay on the line and the color sensors to accurately track the line.
    - We could have two color sensors right next to each other an inch off the ground in order to track the line well but not a big range.
    - We could have one color sensor to track the line and turn until it gets back onto the line.
    - We could have two color sensors right next to each other but farther off the ground for a higher detecting range

- **Picking Object Up:** We needed the robot to be able to lift objects off the ground
    - We could have the tracks in front of the robot so it rides onto the top of it into a holder.
    - We could have a elevator / forklift like system that scoops the object up from the bottom or top and raises it.

- **Determining Contents:** We need the robot to be able to determine the contents of a boxed based on its weights.
    - We could calculate weight based on the motor power used to lift our container. This would give us a different value for every weight then we correspond it with a different possible content.
    - We could calculate weight by putting it on a “scale” and weighing pressure than running a code to give us the value.

Decision Matrix

Movement /Turning	Turning Ability	Smoothness	Code Difficulty	**Line Sensing**	Detecting Accuracy	Detecting Range	Design Implications
4 Wheels	Bad	Bad	Good	1 Color Sensor	Medium	Medium	Low
2 Wheels & Ball Bearing	Good	Good	Medium	2 Color Sensors (1 Inch above)	High	Medium	Medium
3 Wheels (Two front ones back)	Medium	Bad	Medium	2 Color Sensors (2+ Inch above)	Low	High	Medium

Picking Up System	Ability To Lift	Ease Of Code	Complexity Of Design	**Determining Contents**	Design Implications	Code	Ease of movement
Like a Forklift	High	High	High	Calculating Based On Motor Power Used	Low (already have lifting mechanism)	High	Med
Angled Conveyor Belt	Medium	Medium	Low	Scale Method	High	High	Low

Prototyping

Solution To The Needs Above:

- **Movement & Turning:** Of course we needed our robot to be able to move and turn in order to complete any task. **Solution** We decided to use two wheels in front and a ball bearing in the back due to it easing the turn and allowing for a 360 turn without moving. And for the turning we set up the wheels on an isolated system with an independent code for each so one could stop while the other moves to turn. One motor would turn at 40 while the other is at 0. **(See Figure 0)**

- **Path Follower:** We needed our robot to be able to follow a black line that is either solid or dashed, and changes directions. **Solution**: We decided to put two color sensors next to each other, so the robot never loses track of where the line is and if a line is not detected we programmed it to turn until it sees a line. **(Figure 0 & 1: old, new designs | Figure 2 & 3: new, old codes)**

- **Picking Up Object:** We need our robot to be able to pick up and object when needed to complete a task. **Solution:** We decided to make a forklift type system, we have a motor that moves a gear that’s attached to a rigid piece which then raises “forks” up or down which then picks up the object up from the top with the forks then lift it up from there. **(Figure 4: design | Figure 5: pick up Code | Figure 6: drop off code)**

- **Determining Contents:** We need our robot to be able to determine what’s in the box based on its weight. **Solution:** We decided to use the motor power solution. As our robot lifts different weights, there is going be a different strain on the motor. From there we can calculate how much weight is hanging on it then correlate it with a content in the container. **(Figure 7: Code | Figure 4: Design)**

Figure 0: Old Path Following

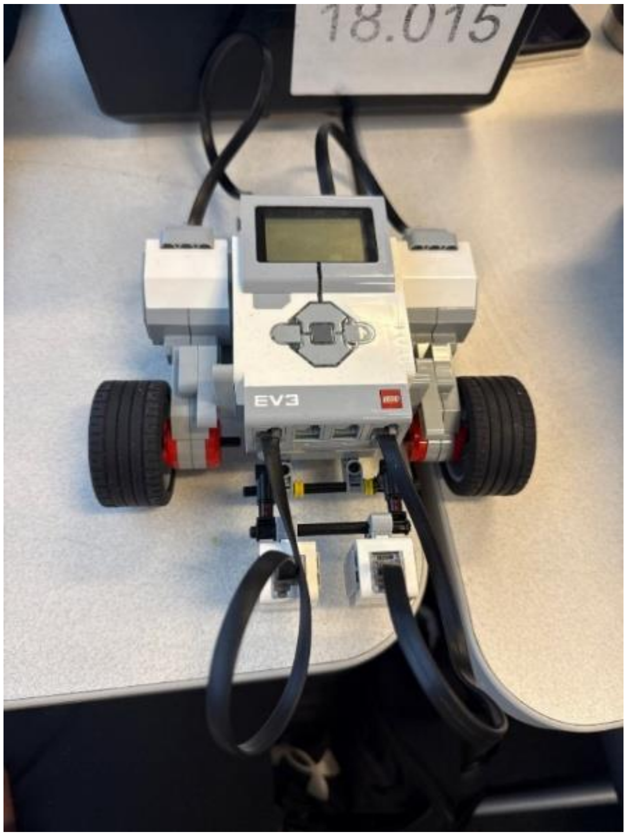


Figure 1: New Path Following

Figure 2: New Path Following Code

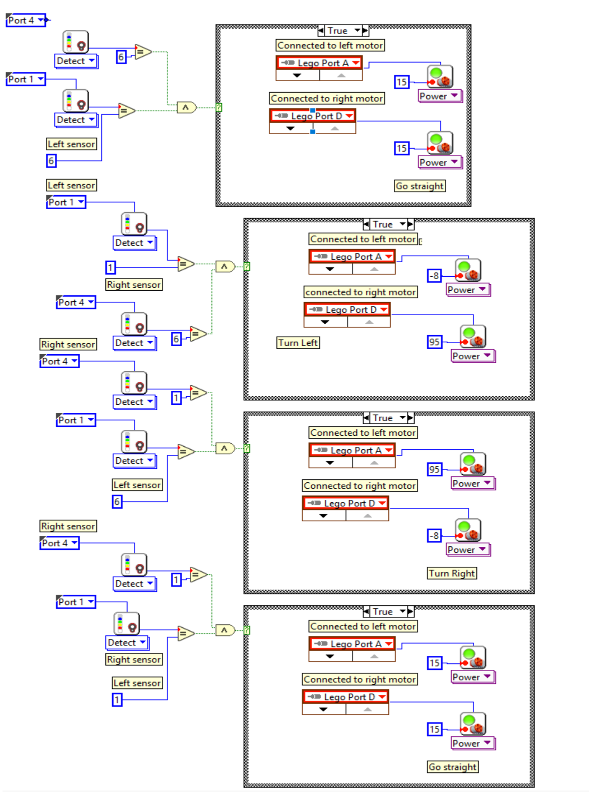


Figure 3: Old Path Following Code

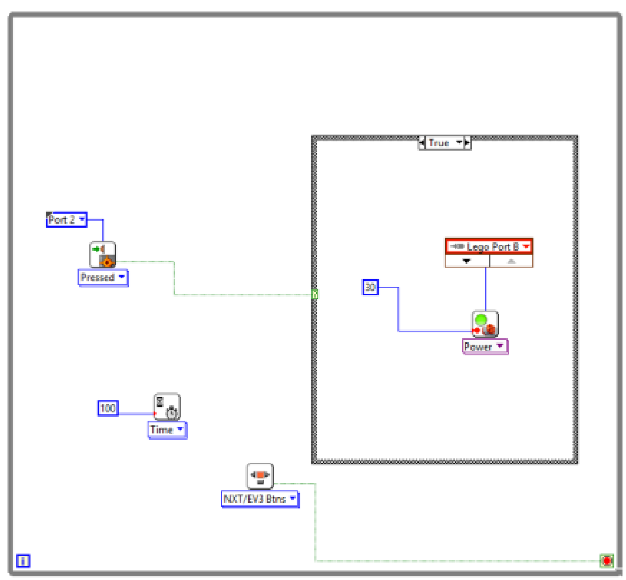


Figure 4: Picking Up Design

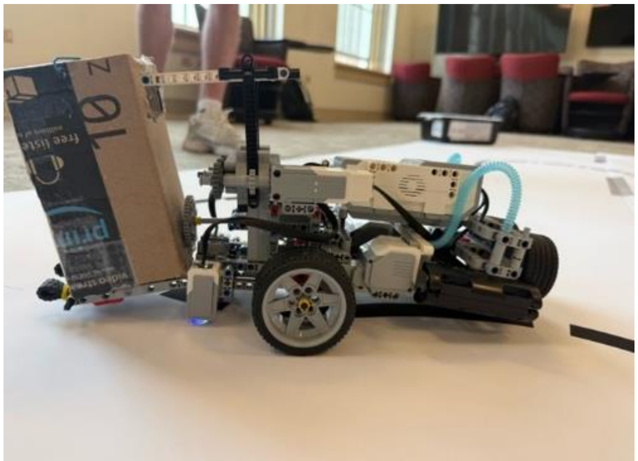


Figure 5: Pick Up Code

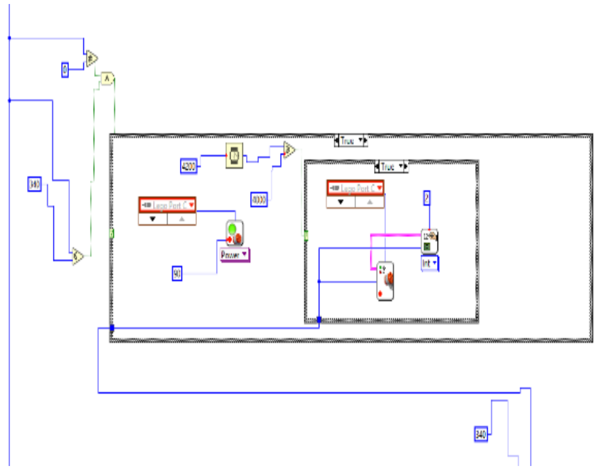


Figure 6: Drop Off Code

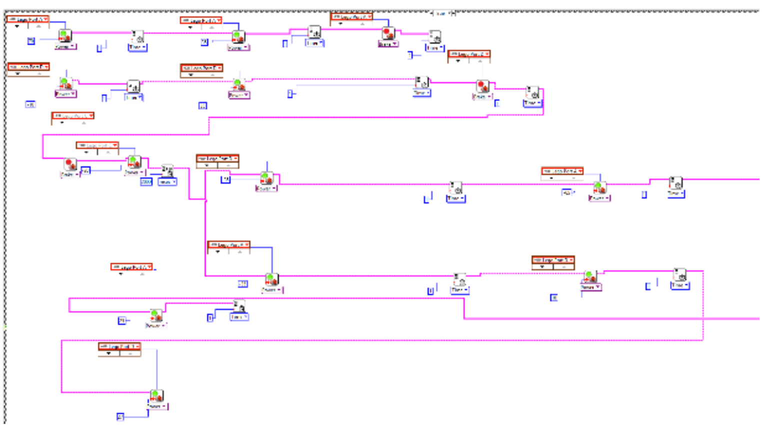
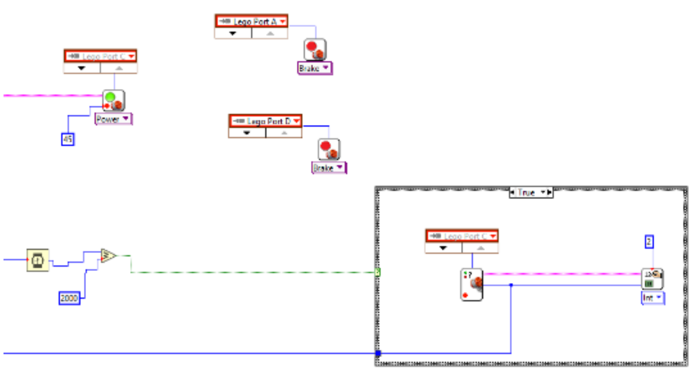


Figure 7: Determing Contents Code



Testing

Testing Plan:

Test Number Assign a number to your test to make referencing easier.	**Criteria/Specifications** List the criteria and specifications you are assessing with the current test.	**Description of Test Procedure** Describe how the test will be accomplished. List what will be done, how many times the test will be repeated, what equipment is needed, etc.	**Data to be Collected** Describe what data will be collected and how it will be obtained.	**Summary of Results** Summarize how you would analyze the data collected to determine the results of the test, specifically stating how the data and analysis relates to the design's performance against the criteria/specifications.
1.1	**Drive Straight:** Robot must maintain a straight path for 3 meter with less than 10cm deviation	We are going to place the robot on a flat surface then have it drive for 3 meter and see if it went straight or not. We are going to do this 3 times.	Deviation from the center line, total distance traveled, and time.	If our robot drives straight (within 10cm) for 3 meters, we can conclude that our robot can drive straight over a given distance.
2.1	**Turning Test:** Robot must be able to turn smoothly onto turns up to a 60 degree angle withing 5 degrees	We are going to place the robot on a flat surface and run a code which causes one motor to stop and one motor to go. We are going to measure how far the robot was able to go with a protractor. We are going do this 3 times	Measure the angle of the turn in degrees.	If our robot turned on a 35-degree turn and successfully turned within 5-degrees, we can say that our motors can successfully turn to a given angle.
3.1	**Range of Absorbance:** The color sensors must absorb a range of light which determines the line to follow.	We are going to alter the range of absorbance to see if the robot's motors continue to run while following the line.	How many times does the robot successfully follow the path and how many times does it fail.	If the robot continues to follow a line with the adjusted range of absorbance, then we can say that the color sensors can successfully observe a line within the range of absorbance.
4.1	**Line Following**: Robot must navigate a path with a curve into it without losing the line	We are going to place the robot on a black line with a curve in it and run the program until it is able to do it flawlessly.	How many times does the robot successfully follow the path and how many times does it fail.	If the robot followed the path successfully during each trial, we could conclude that our robot will successfully follow a path with a curve on it.
4.2	**Dashed Line Following:** Robot must navigate a dashed black line with curves in it without losing the line	We are going to place the robot on a dashed line with a curve in it and run the program until it is able to do it flawlessly.	How many times does the robot successfully follow the path and how many times does it fail.	If the robot followed the dashed path successfully multiple times, we could conclude that our robot can successfully follow a dashed line.
5.1	**Lifting mechanism:** Can the robot successfully pick the box with a weight of 150 grams?	Put weight on the lifting mechanism in 50g increments and attempt to lift until failure	How much weight can the lifting mechanism hold before it doesn't lift anymore.	If the robot can carry a weight greater or equal to 150 grams, we know that our robot can pick up a crate of a weight from 0-150 grams.

5.2	**Determining Content** Robot must be able to identify the object based on motor power	Run the lift multiple times with different weights then see if it calculates it right. Repeat 5 times.	Motor power for different weights.	If our robot correctly calculates the weight and determines the contents correctly than we know our code and design works and it passes the test.

We plan on testing the following functions: Driving straight, ability to turn, range of absorbance, solid/dashed line following, lifting mechanism, and finally if it will determine contents correctly. We will use **(1)** Description Of Test **(2)** Data to be collected **(3)** Summary

Prior to Subtask 1 (2/17) ---------

<u>1.1 Straight Drive Test: </u>(1) We will test if our robot is capable to drive straight for three meters because if it can't, it will make every task difficult in the future. We will use apples measure feature to calculate the distance then we would find out if it deviated from the path or not. (<10cm deviation allowed) We will do this 3 times. **(2)** Data that is collected is going be deviation from the line, and total distance. **(3)** This will tell us if our robot can successfully go straight, smoothly or not.

	Trial 1	Trial 2	Trial 3	Mean
Deviation	5cm	2cm	6cm	6cm
Total Distance	3m	3m	3m	3m

Conclusion: In conclusion we found that our robot is capable of driving in a straight line just by itself within a 10 cm deviation. This solves the criteria for driving straight.

<u>2.1 Turning Test: </u>(1) We will test if the robot is able to smoothly turn up to 60 degrees. We will do this by taking a angle measuring device either on apple or a protractor and seeing if it can do a 180 or turn up to at least 60 degrees within a 5-degree error on a flat surface. We will do this test 3 times. **(2)** Data to be collected is the measure of the angle turned in degrees. **(3)** This will tell us if our robot can successfully turn or not.

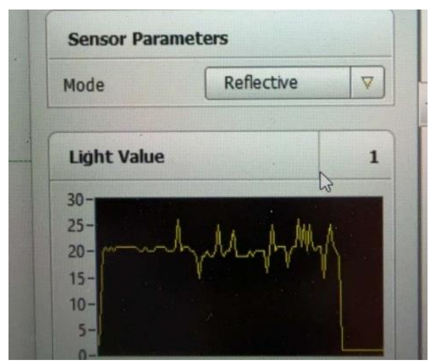
	Trial 1	Trial 2	Trial 3	Mean
Angle of turn (degrees)	357	360	363	360

Conclusion: In conclusion we found that our robot can turn in a full circle if needed too within a 5-degree precision.

This solves the criteria for turning.

<u>3.1 Range of Absorbance Test:</u> (1) We are going to alter the range of absorbance until the robot follows the line flawlessly. This is a pass or fail test and we do not need any addition material besides a line. **(2)** Data to be collected is if the robot tracked the absorbance values correctly and stayed on the line. **(3)** This will tell us if our robot will pick up the line correctly, if it does we can move onto making it stay on the line.

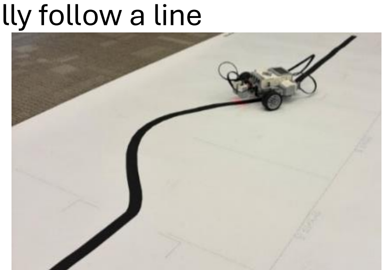
	Trial 1	Trial 2	Trial 3
Successful Run	Yes	Yes	Yes



Conclusion: In conclusion we found that our robot could successfully detect the right absorbance values within a 5-value precision (calculated thru MATLAB display values). This meets the criteria for sensors working.

<u>4.1 Line Following Test (Subtask 1):</u> (1) We are going to place the robot on a solid black line with a curve in it and run our line following code. We will do this as many times as needed until it passes, this is a pass fail test. We will need a track to test than we will test at the Subtask also. **(2)** Data to be collected is if the robot completed the course or not. **(3)** This will tell us If our robot will pass the first subtask and also successfully follow a line

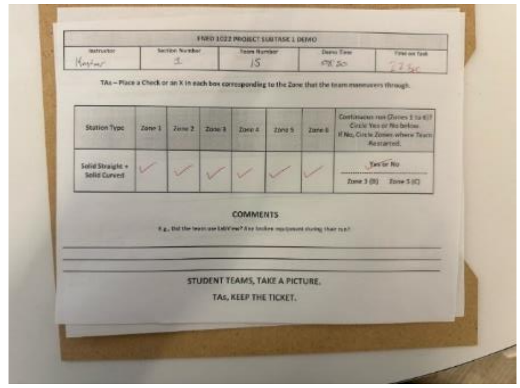
	Trial 1	Trial 2	Subtask 1
Successful Run	Yes	Yes	Yes



Conclusion: In conclusion we found that our robot could successfully follow a solid line

(yes or no task, no certain requirements). This

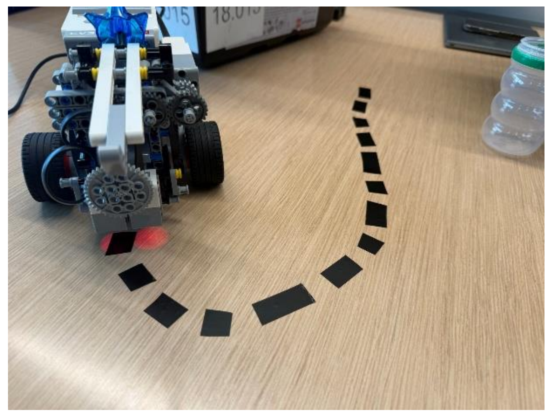
meets the criteria for tracking a solid/dashed line.



Prior to Final Demo (4/15) ---------

<u>4.2 Dashed Line Following Test:</u> (1) Mostly the same as the previous, we are going to place the robot on a dashed black line with a curve in it and run our line following code. We will do this as many times as needed until it passes, this is a pass fail test. We will need a track to test on. **(2)** Data to be collected is if the robot completed the course or not. **(3)** This will tell us If our robot is capable of a more advanced track and will tell us if it can follow a dashed line or not.

	Trial 1	Trial 2	Trial 3
Successful Run	Yes	Yes	Yes



Conclusion: In conclusion we found that our robot

could successfully follow a dashed line

(yes or no task no certain requirements). This meets

the criteria for tracking a solid/dashed line.

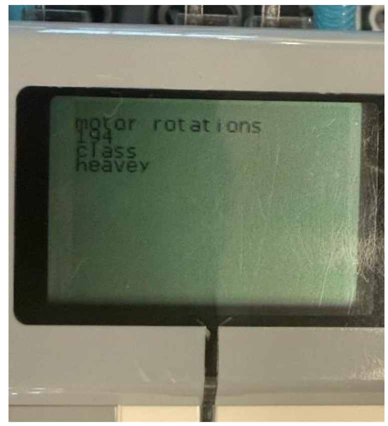
<u>5.1 Lifting Mechanism Test:</u> (1) We will put different weights in our container and have the lifting mechanism lift it from 0 – 150g in 50g increments. We will be using change to weigh it down and will repeat once for every increment. **(2)** We will collect whether or not the lifting mechanism made it to the top or if it could not lift it. **(3)** This will tell us if our robot is capable of picking up object or not.

Pound	Result
~0g	Yes
~50g	Yes
~100g	Yes
~150g	Yes

Conclusion: In conclusion we found that our robot could successfully lift objects up to 150 grams. This meets the criteria for being able to pick objects up.

<u>5.2 Determining Content Test:</u> (1) We will run the lifting system multiple times with different weights and in order to make sure they correspond to the right content that's in the container, we will do this 5 times. **(2)** We will collect motor power for different weights so all we need is MATLAB and the robot. **(3)**

This will tell us if our robot can successfully determine contents based on weight levels or not.

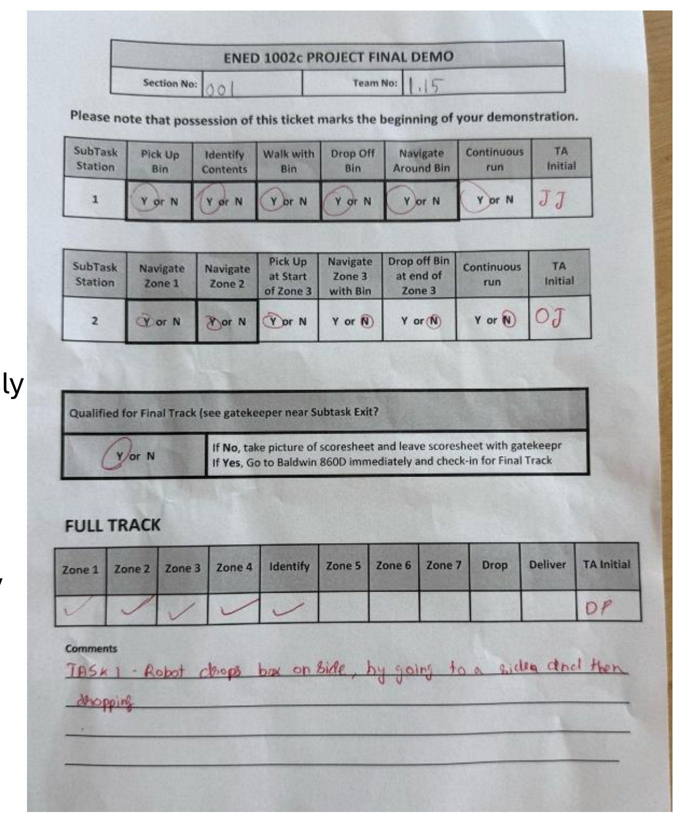


Weight	Item
0g	Light
50g	Light
100g	Heavy
150g	Heavy

Conclusion: In conclusion we found that our robot could successfully determine if the weight was heavy or not. This meets the criteria of determining contents.

Results – Final Demo

In reflection we tested with a different heavy weight then used in the final demo. This caused their heavy weight to apply pressure differently and our color sensors to touch the ground. We could complete light flawlessly but our heavy would slowly stop working due to the bottom getting closer and closer to the ground. Our robot would of worked flawlessly if we fixed that one small issue.



ENED 1002c PROJECT FINAL DEMO

Section No: 001 Team No: 1.15

Please note that possession of this ticket marks the beginning of your demonstration.

SubTask Station	Pick Up Bin	Identify Contents	Walk with Bin	Drop Off Bin	Navigate Around Bin	Continuous run	TA Initial
1	Y or N	Y or N	Y or N	Y or N	Y or N	Y or N	JJ

SubTask Station	Navigate Zone 1	Navigate Zone 2	Pick Up at Start of Zone 3	Navigate Zone 3 with Bin	Drop off Bin at end of Zone 3	Continuous run	TA Initial
2	Y or N	Y or N	Y or N	Y or N	Y or N	Y or N	OJ

Qualified for Final Track (see gatekeeper near Subtask Exit?	
Y or N	If **No**, take picture of scoresheet and leave scoresheet with gatekeepr If **Yes**, Go to Baldwin 860D immediately and check-in for Final Track

FULL TRACK

Zone 1	Zone 2	Zone 3	Zone 4	Identify	Zone 5	Zone 6	Zone 7	Drop	Deliver	TA Initial
✓	✓	✓	✓	✓						DP

Comments

TASK 1 - Robot drops box on Side, by going to a sidea and then dropping